Explainable Machine Learning for Chronic Kidney Disease Prediction

SHAP-based feature attribution on the UCI CKD dataset

Vedank Singh | Reg. No. 240962043

B.Tech. CSE (AI & ML), V Semester, Section B

Guides: Dr. Narendra V G (Professor) and Ms Vibha (Assistant Professor - Senior Scale)

School of Computer Engineering, Manipal Institute of Technology | Academic Year 2026–27

Why CKD prediction needs explanations

788 million

adults had CKD in 2023 (age-standardised prevalence 14.2%), GBD 2023 Study

Often silent early
Routine blood and urine tests can flag likely CKD before symptoms appear.

Black boxes are hard to trust
A clinician needs to see why a model flags a patient. SHAP attributes each prediction to the inputs.

Objectives

1. Acquire, clean and preprocess the UCI CKD data without leakage: missing values, encoding, scaling, class imbalance
2. Tune and compare Logistic Regression, SVM, Random Forest and XGBoost on accuracy, precision, recall, F1 and ROC-AUC
3. Explain the chosen model with SHAP, globally and per patient, and compare its key features with clinical literature

The data: 400 patients, 24 routine clinical inputs

400
records (UCI repository, id 336)

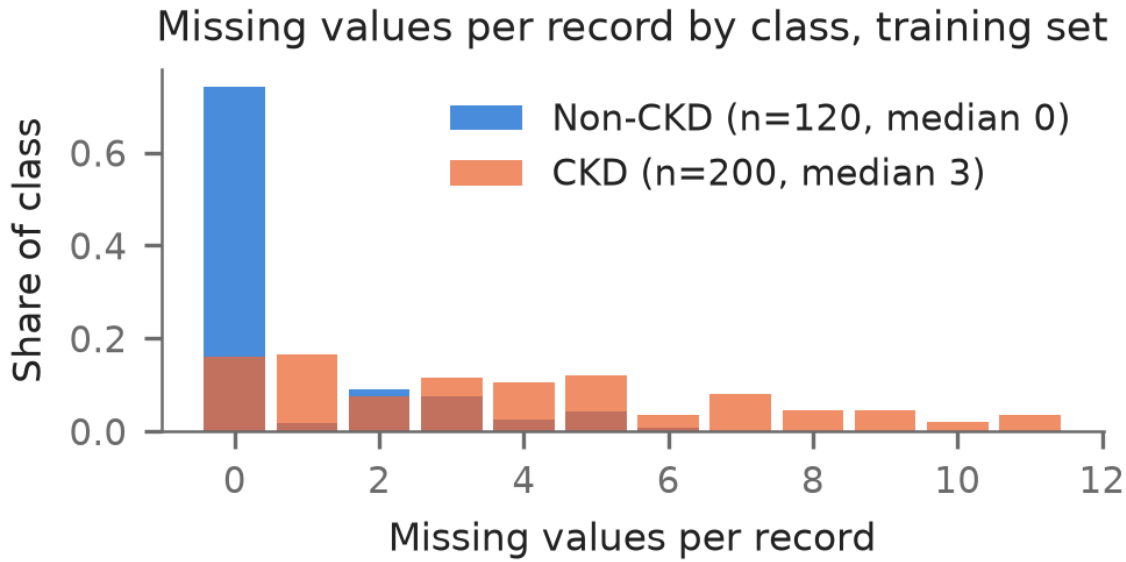
250 / 150
CKD / non-CKD

24
inputs: 11 numeric, 3 ordinal, 10 binary

10.6%
of values missing

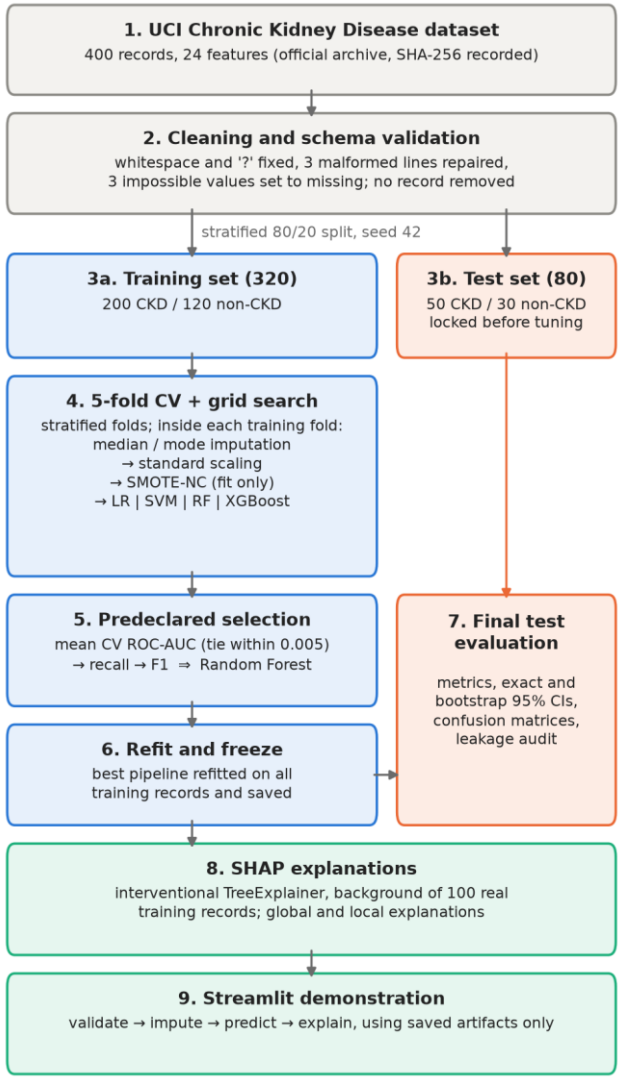
Cleaning: nothing deleted

- Official archive cached with SHA-256; matches UCI's CSV cell for cell
- 3 malformed lines and 16 stray tab/space cells repaired
- 3 impossible values set to missing (sodium 4.5; potassium 39 and 47 mEq/L)
- No duplicate patient records; 80-record test set locked before tuning



In training data, CKD records miss far more values, and no non-CKD record shows any abnormal finding: the classes are unusually easy to separate (creatinine median 2.3 vs 0.9 mg/dL).

Pipeline designed so the test set stays untouched



How leakage is prevented

- ✓ Test set (80 records) split off and locked, with a checksum, before any tuning
- ✓ Imputation, scaling and SMOTE-NC are refitted inside every training fold
- ✓ SMOTE-NC runs only during fitting: validation and test records are never resampled
- ✓ The model is chosen from training-set cross-validation; the test set is used once
- ✓ Record ID and target are excluded from the predictors
- ✓ Automated audit: 11 / 11 checks passed; shuffled labels give CV ROC-AUC 0.505

Four models, one selection rule fixed in advance

Logistic Regression
L2 penalty, C = 0.1

SVM
RBF kernel, C = 1, gamma = scale

Random Forest (selected)
300 trees, full depth, sqrt features/split

XGBoost
100 trees, depth 2, learning rate 0.05

5-fold CV (mean ± SD)	ROC-AUC	Recall	F1-score
Logistic Regression	0.999 ± 0.001	0.960 ± 0.041	0.979 ± 0.021
SVM	1.000 ± 0.000	0.985 ± 0.030	0.992 ± 0.016
Random Forest	**0.999 ± 0.001**	**1.000 ± 0.000**	**0.995 ± 0.006**
XGBoost	0.999 ± 0.002	0.985 ± 0.012	0.985 ± 0.012

Rule (written before any score)

1. Highest mean CV ROC-AUC wins
2. Within 0.005: tied, ranked by recall, then F1

All four tied (0.9988–1.0000): Random Forest wins on recall (1.000)

The synopsis proposed picking the best test score; selecting on training CV keeps the test estimate unbiased.

Untouched test set: near-perfect, with honest uncertainty

80 / 80

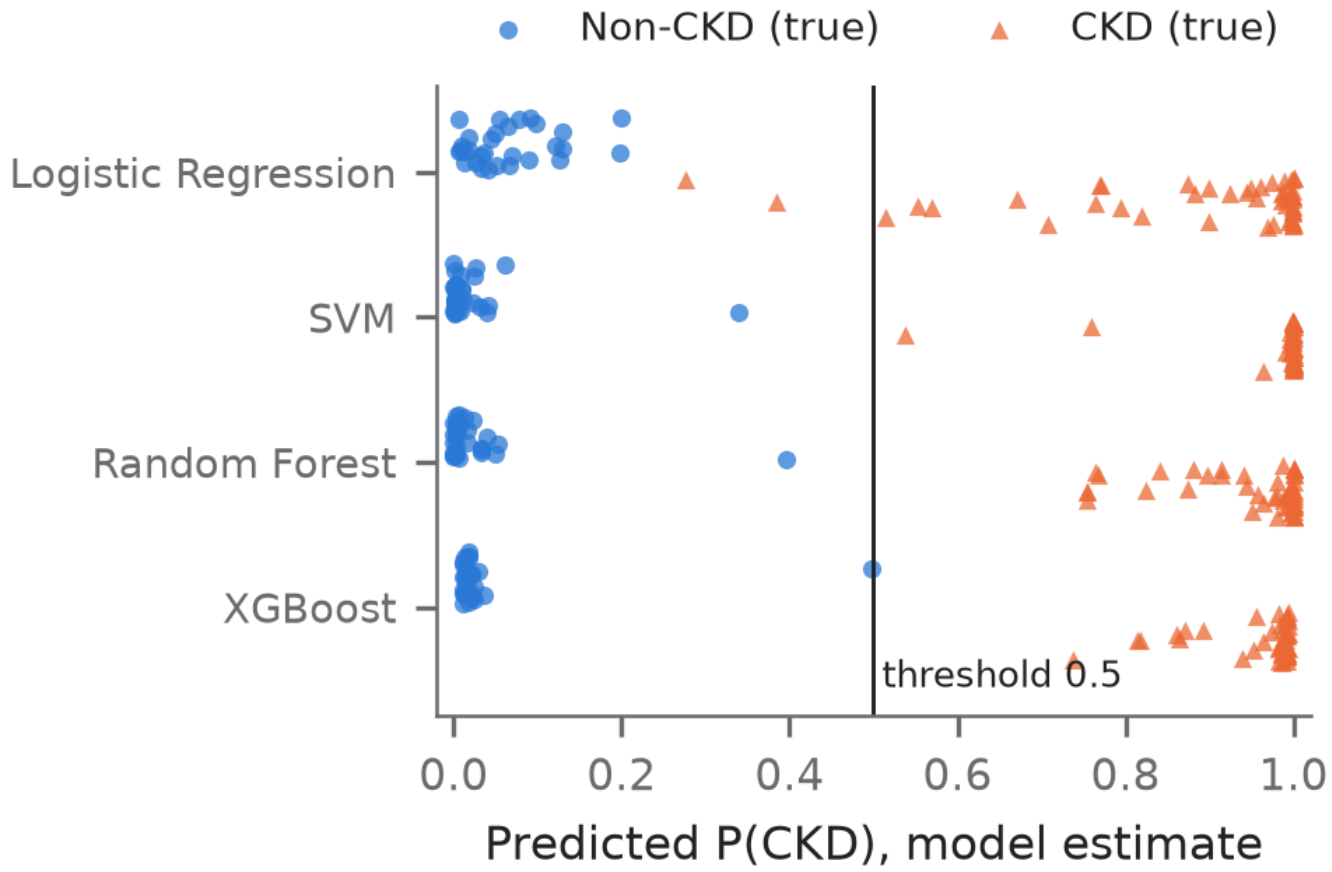
test records correct for Random Forest (also SVM and XGBoost)

0.955–1.000

exact 95% CI for Random Forest accuracy

2

CKD cases missed by Logistic Regression (no FP)



Perfect accuracy hides a near miss: non-CKD record 324 scored 0.397 (Random Forest) and 0.497 (XGBoost), just below the 0.5 threshold.

Checks that keep the results honest

No gain

SMOTE-NC vs no resampling

Same folds and settings: recall and F1 changed by at most 0.010, within one fold SD. Kept as specified, no benefit claimed.

0.864

Missingness alone

CV ROC-AUC from which values are missing, ignoring the values: testing practice carries class signal.

0.505

Shuffled labels

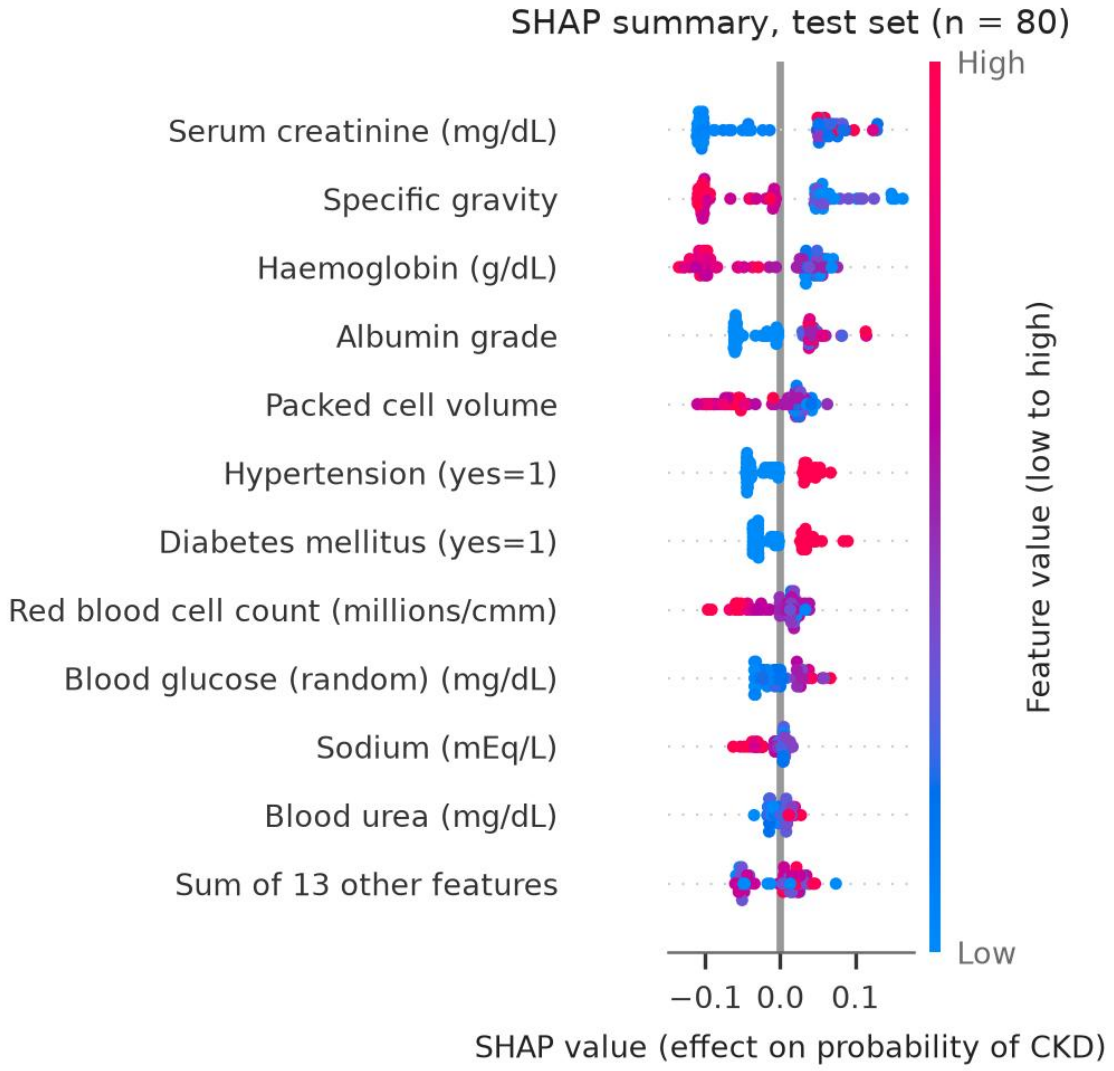
CV ROC-AUC after permuting the labels: chance level, so the pipeline has no hidden leak.

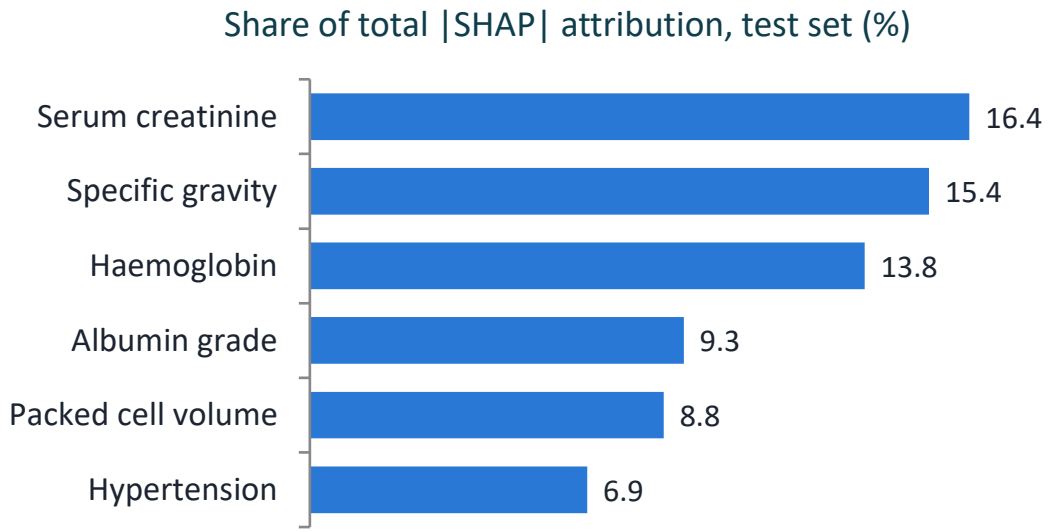
0%

Abnormal findings in non-CKD

No non-CKD training record has hypertension, diabetes, anaemia or any other abnormal finding.

What drives the predictions: global SHAP

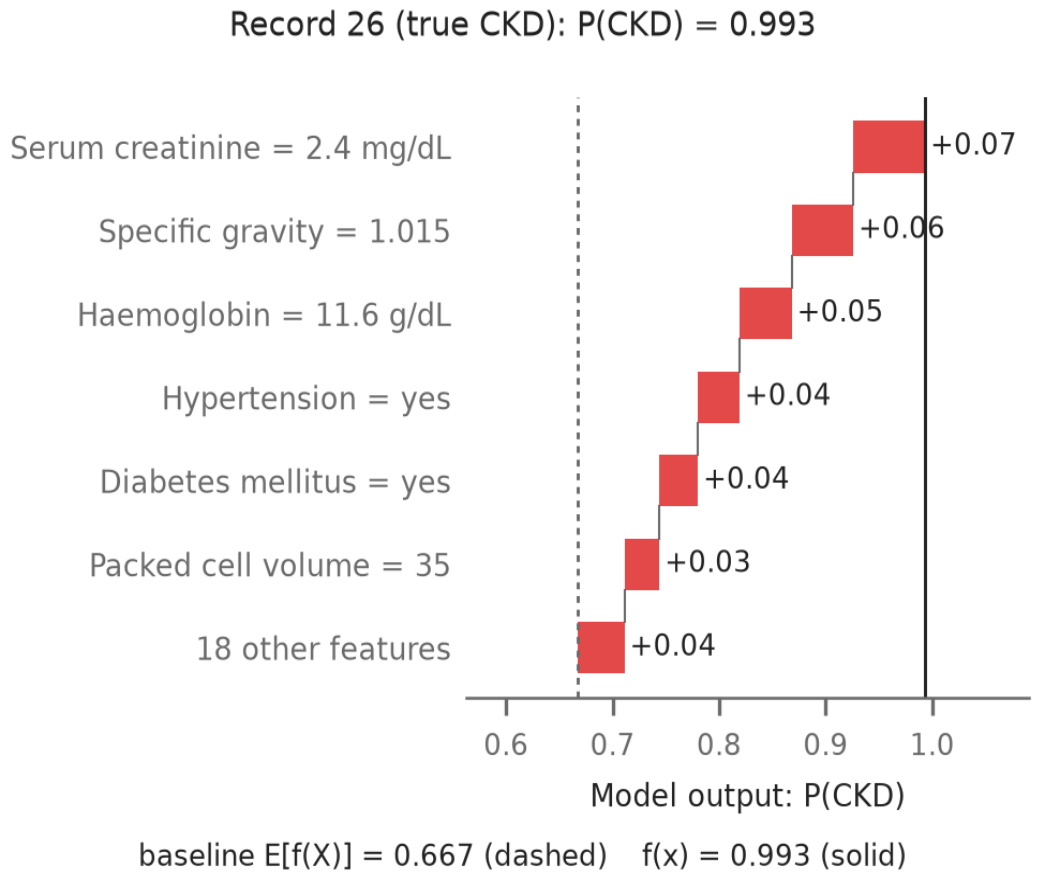


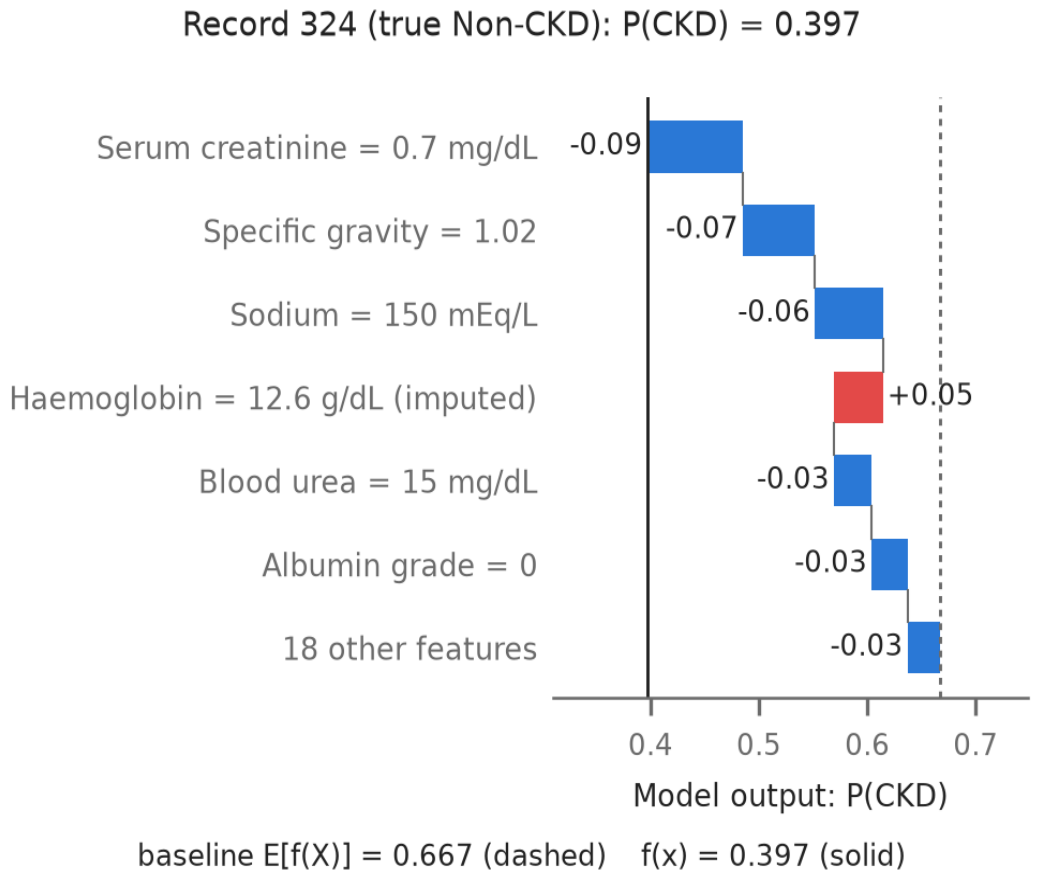


- Creatinine: basis of eGFR, which defines CKD
- Albumin: marker of kidney damage
- Haemoglobin, packed cell volume: anaemia, common as CKD advances
- Hypertension, diabetes: leading CKD risk factors
- Specific gravity: not a diagnostic criterion; matches earlier SHAP work on this dataset

Ranking stable between test and training records (Spearman 0.995). SHAP explains the model, not the disease.

Why one patient got their prediction: local SHAP

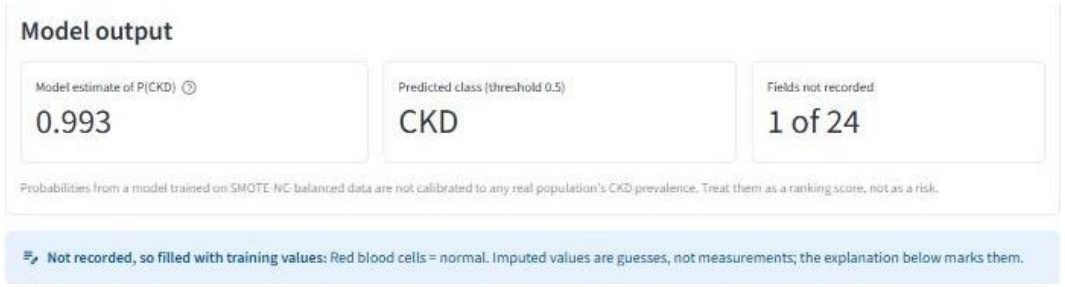




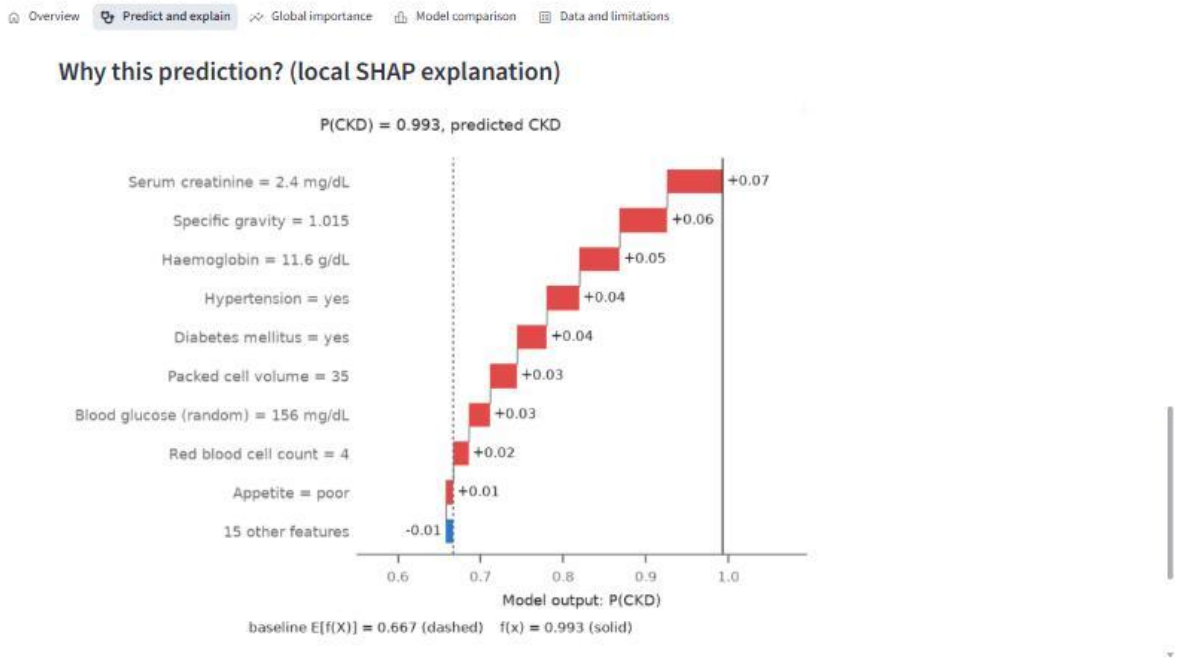
Record 26 (true CKD), P = 0.993
High creatinine, low specific gravity and low haemoglobin push the output up from the 0.667 baseline.

Record 324 (true non-CKD), P = 0.397
Normal creatinine pushes down, but missing haemoglobin, imputed at the training median 12.6 g/dL, pushes up.

Demonstration: a local Streamlit prototype



1. Validate inputs against the schema
2. Impute blanks and list them
3. Predict P(CKD) with the saved model
4. Explain the prediction with SHAP
5. Check baseline + contributions = output



Screenshots from the running app (dataset record 26). It loads saved artifacts only: about 46 ms per prediction and 82 ms per explanation. Every page states that it is an educational prototype, not a diagnostic tool, with no treatment advice.

Conclusion, limitations and next steps

What we found

- A reproducible, leakage-audited pipeline for four classifiers
- Random Forest chosen by a predeclared CV rule; 80/80 test records correct
- SHAP: creatinine, specific gravity, haemoglobin, albumin, packed cell volume
- Oversampling gave no consistent benefit

Limits

- 400 single-centre records; healthy-looking controls
- 80 test records: wide confidence intervals
- No eGFR or repeat tests; missingness is informative
- Probabilities uncalibrated; SHAP is not causal
- No external validation: not clinically validated

Next steps: external validation on larger cohorts with eGFR-based labels, calibration, explicit missingness modelling, and explanation stability across models.